P7 TERM ONE LESSON NOTES

TOPIC: **CLIMATE OF AFRICA**

SUBTOPICS: **Climatic Regions in Africa**

Characteristics of different Climatic regions

Factors that influence climate of Africa

Influence of climate on human activities

Influence of human activities on climate

SESSION	COMP	CONTENT	METHODS	RELEVANCE	LESSON ACTIVITY	SCENARIOS
1.	Discusses how elements of weather helps one to know the types of climate in Africa	**INTRODUCTION TO WEATHER** **Weather** is the state of the atmosphere of a place at a given time. **Meteorology** is the scientific study of weather. **A meteorologist** is a scientist who studies weather. Elements of weather are measured and recorded at a weather station/ meteorological centre. The biggest meteorological centre/ weather station in Uganda is Entebbe meteorological centre. **Weather forecasting** is the telling of the expected future weather conditions of an area. **Importance of weather forecasting.** It helps farmers to plan well their farm activities. e.g. Planting, harvesting etc.	Guided discovery Observation Brainstorming Question and answer	To help the learner understand which activities to do depending on weather conditions To help the learner understand safety and deserter preparedness	Discussing how elements of weather helps one to know the types of climate in Africa	Learner going to school Entebbe Meteorological Centre Farmers Travellers

		It helps travellers to prepare for their journeys. It helps pilots and sailors to avoid air and water accidents. **Main aspects of climate** Rainfall, temperature **Types and conditions of weather** Rainfall Humidity Cloud cover Sunshine Temperature Wind Air pressure **Importance and dangers of various weather elements**	Guided group discussion	To help learners understand the environment around him/her		Pilots & Sailors The school assembly The sports day Drying clothes Dusty road Boda boda riders

Type of weather	Condition of weather
Rainy weather	Rainy
Windy weather	Windy
Cloudy weather	Cloudy
Sunny weather	Sunny

Element	Importance and dangers

			Wind	<u>Wind</u> is air in motion. Wind is caused by the difference in atmospheric pressure. Wind blows from the areas of high atmospheric pressure to areas of low pressure. <u>Importance</u> It is used in winnowing. Wind drives away bad smell. It dries people's clothes. It helps in sailing boats on large water bodies. It helps in pollination. It helps in flying kites and balloons. It can be turned into power by wind mills. Wind mills are mainly used in Karamoja to pump water. <u>Dangers.</u> Strong wind destroys farmers' crops. Wind causes soil erosion. It pollutes the environment by raising dust. It drives away clouds that would bring rainfall.				Market day Cooking outside Village football match Shadows on the ground Observing your breath Roof noise at night

			Strong wind blows off roofs of houses. Strong wind makes water vessels capsize on water bodies. Dry winds cause desert conditions in areas where they blow to.				
			Clouds	These are solid drops of rain in the atmosphere. <u>Importance</u> Clouds protect us from direct sun rays. Some clouds give us rain eg. Nimbus clouds. Clouds regulate world temperatures. Clouds keep the earth warm at night.			

			Rainfall	Rainfall is the amount of rain received in a particular area. Areas that receive plenty of rainfall are usually densely populated while those that receive unreliable rainfall usually have very few people (are sparsely populated) Types of rainfall Convectional rainfall (mostly received around forests and large water bodies) Relief/ Orographic rainfall (received mostly in highland areas) Cyclonic/ frontal rainfall (received mostly in plateau areas/ plains) Importance Rain is a natural source of water in the environment. It helps plants to grow well. It helps to reduce dust in the environment. It helps to cool the environment. It increases the volume of water in water bodies. Rain water softens the soil for cultivation.				

				<u>Dangers</u> Too much rainfall causes floods. It causes landslides in mountainous areas which lead to displacement of people, death of people and animals, destruction of property etc. It leads to soil erosion. Heavy rainfall destroys farmers' crops. It makes murram roads muddy and slippery.				

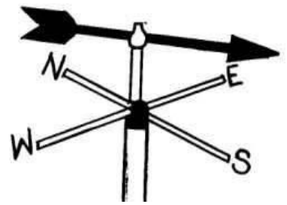
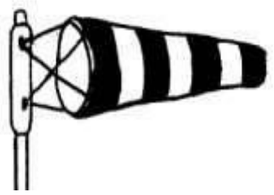
			Sunshine	The sun is the major source of light in the environment. Africa receives abundant sunshine throughout the year because the sun is overhead Africa throughout the year. Sunshine is tapped and utilized by many people in Africa in form of solar energy. <u>Importance</u> It is used in preserving food. It is a source of Vitamin D. It dries farmers' crop harvests. It is used in generation of solar energy. It helps in drying clothes. <u>Dangers</u> Strong sunshine leads to drought. Direct sun rays cause skin cancer. It leads to global warming. It dries up pasture for animals. It causes dust in the environment.				

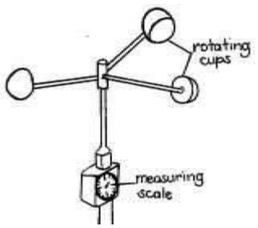
	It dries crops in the garden before they are ready for harvesting.

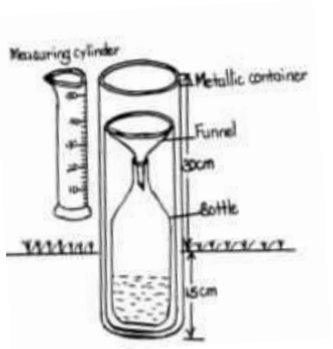
WEATHER INSTRUMENTS

These are tools used in measuring and recording atmospheric conditions.

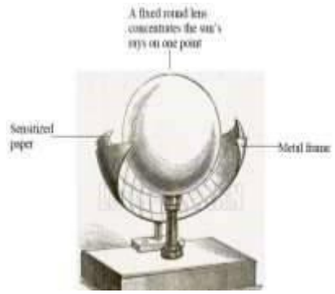
Weather instrument	Importance
Wind vane / a weather cock	It shows the direction of wind. A wind vane is placed in an open place or on top of buildings to prevent wind obstruction.

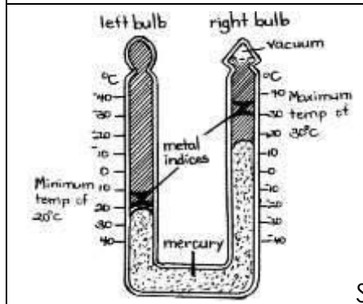
			Note:-An arrow of a wind vane usually points to the direction from which wind is blowing.				
		A  Windsock	It is used to determine the strength of wind. It's usually found at the airports, air fields, at chemical plants etc. Note:-A wind sock is not among the instruments of weather found at a weather station.				

			 Anemometer	An	It measures the speed of wind. It has cups which trap wind and rotate as wind blows into them. The speed of wind is measured in kilometres per hour (km/hr)				

			A Rain gauge  	It is used to measure the amount of rainfall received in an area. It is always placed at least 30cm above the ground to prevent splashes and flowing water on the ground from entering the rain gauge. It should be placed 15cm below the ground to prevent the evaporation of water collected and to make it firm. It is placed in an open flat place to prevent obstruction of rain drops in order to take accurate measurements. **Importance of different parts of a rain gauge.** The funnel-it directs water into the bottle. Water bottle-it collects the amount of rain water received. Measuring cylinder-it is used for measuring the amount of water collected in the bottle.				

				Note:-Rainfall is measured in millimetres in order to know the depth of rain water into the soil.				
			A ceilometer	It is used to measure the height and thickness of clouds.				

			 A Barometer	It is used to measure air pressure. Air pressure is measured in millibars. Measuring air pressure enables experts to predict storms.				
			 Sunshine recorder	A It shows the duration of sunshine in a day.				

		 Hygrometer	It is used to measure humidity. Humidity is the amount of water vapour in the atmosphere.				
		 Six's thermometer	It measures the highest and lowest temperatures of the day.				
		A STEVENSON SCREEN.					

		This is a wooden box with louvres on metallic stands in which delicate weather instruments are kept at a weather station. It protects weather instruments from rainfall and direct heat. The Stevenson screen is made of louvres to allow free air circulation in the box. It is painted white to enable it reflect sunlight. It is raised 1.5m above the ground to keep it free from splash water. It is made of wood to prevent it from absorbing heat.  **Weather instruments kept in a Stevenson screen** Barometer Hygrometer Six's thermometer (Maximum and thermometer). **Note:** The above instruments are kept in a Stevenson screen in order to protect them from destruction since they are delicate. The Ministry of Water and Environment is responsible for monitoring weather and climate, and issuing advice to the public on weather conditions in Uganda.					

2.	Identifiesthe climatic regions of Africa and their characteristics Draws the map of Africa showing climatic regions Draws graphs, charts of tables showing different climatic regions Explains the different characteristics of each type of climatic region	**CLIMATE OF AFRICA.** Climate is the average weather condition of a place recorded for a long time. Climatology is the scientific study of climate. A climatologist is a scientist who studies climate. A climatic region is a large area that experiences the same / similar weather patterns. The climatic regions of Africa include the following; Equatorial climatic zone. Tropical (Savannah) climatic zone Semi-desert/ Semi-arid climatic zone Desert / Arid climatic zone Mediterranean climatic zone. Temperate climatic zone. Mountain / Montane climatic zone **MAP SHOWING CLIMATIC REGIONS/ZONES IN AFRICA** **EQUATORIAL CLIMATE** It is described as hot and wet throughout the year. This type of climate is experienced in areas lying between 5ºN and 5ºS of the equator. The high rate of evaporation and transpiration causes heavy rainfall in this region (usually above 1750mm)	Guided discovery Observation Brainstorming Question and answer Guided group discussion	To help the learners appreciate different climatic regions of Africa To help the learner understand why people live and work differently To help learners understand how to respect and care for the environment To help leaners develop map reading skills	Identifying the climatic regions of Africa and their characteristics Drawing the map of Africa showing climatic regions Drawing graphs, charts of tables showing different climatic regions Explaining the different	Morning Dew Long Lines at Water Wells During the dry season, many people queue for water at boreholes. Crop Failure Sweating Easily Thick Fog in Early Morning Staying Indoors at Midday Thunderstorms Dust Storm Flooded Gardens Walking Long Distances for Pasture Wearing Sandals Putting on Closed Shoes

		In Uganda, Equatorial climate is mostly experienced on the shores of lake Victoria. **African countries that experience Equatorial climate** Democratic Republic of Congo, Congo Brazzaville, Gabon, Equatorial Guinea, Central African Republic, Uganda-on the shores of lake Victoria **Characteristics of Equatorial climate.** It is hot and wet throughout the year. Convectional rainfall is mostly received. (mainly in the afternoon.) It has no dry month. It has two heavy maximum rainfall seasons. (a double rain maxima during the equinoxes) It experiences very high temperatures daily. (of about 25º daily on average.) **Note:-** People living in Equatorial climatic region build houses with slanting roofs to allow easy flow of rainwater off the roofs/to allow easy down flow of rain water from the roofs. A climatic table showing **Equatorial type of climate** A climatic graph showing **Equatorial type of climate** **Note:-** According to the above table and the graph, the increase in temperatures causes increase in the amount of rainfall received in the Equatorial climatic region.			characteristics of each type of climatic region	Food Shortages Children Playing in the Rain Solar Panels Working Better Wet Clothes Taking Longer to Dry Livestock Drinking More Water Tourists Choosing Hot Beaches

		Economic activities carried out in equatorial climatic region Crop growing Lumbering Tourism **Common crops grown in equatorial climatic region** Oil palm, Rubber, Bananas , Cocoa, Coffee **Note:-** The above crops are commonly grown because they require plenty of rainfall for them to grow well.				Teachers Avoiding Yard Duty in Midday Heat Children Using Mosquito Nets More in Rainy Season Packing for a Trip Farming Season House Construction

						Seasonal Fruits Water Availability Fog in the Highlands Flooded Roads Desert Travel Lake Breeze

						Drought Snow on Mountain Tops Umbrella Habit Dusty Roads Fishing Restrictions Clothing Choice Travel Decision

						Water Shortage Comparing Two African Regions
3.	Identifies Tropical climatic region of Africa and their characteristics Draws the map of Africa showing Tropical climatic regions Draws graphs, charts of tables showing Tropical climatic region Explains the different characteristics of Tropical climatic region	**ii) TROPICAL CLIMATE / SAVANNAH CLIMATE** Tropical climate is described as hot and wet. Tropical type of climate is experienced in most parts of Africa. This is because most parts of Africa lie between the tropics (with in the tropical region). This type of climate is experienced between the Tropic of cancer and the Tropic of Capricorn. Tropical climatic region lies in areas between 5º to 15ºN and 5º to 15ºS of the equator. **African countries that experience Tropical climate** Uganda, Zambia, Zimbabwe, Kenya, Tanzania, Rwanda, Ghana, Nigeria, Burundi, Mali, Malawi **Characteristics of Tropical climate** Rainfall is mainly received when the sun is overhead the equator. It has two wet seasons and two dry seasons.	Guided discovery Observation Brainstorming Question and answer Guided group discussion	To enable leaners understand seasons of the year To enable learners understand vegetation and wildlife To enable learners appreciate Africa's diversity To enable learner explain how tropical	Identifying Tropical climatic region of Africa and their characteristics Drawing the map of Africa showing Tropical climatic regions Drawing graphs, charts of tables	Morning Dew Long Lines at Water Wells Crop Failure Thunderstorms Flooded Gardens Putting on Closed Shoes

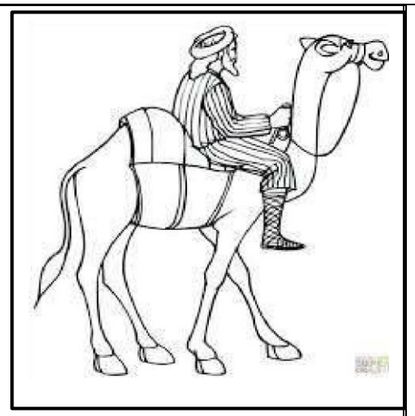
		Rainfall decreases as one moves far from the equator. Tropical type of climate is hot and wet. **The climatic table showing Tropical type of climate** **A climatic graph showing Tropical type of climate** **Note:-** According to the above table and the graph, the increase in temperatures causes increase in the amount of rainfall received in the Tropical climatic region. **Economic activities commonly carried out in Tropical climatic region** Crop growing, Bee keeping, Lumbering, Tourism, Pastoralism **Common crops grown in Tropical climatic region** Maize, Sorghum, Beans, Ground nuts, Fruits like; jack fruits, oranges, mangoes, guavas	Picture interpretation	climate influences human activity	showing Tropical climatic region Explaining the different characteristics of Tropical climatic region	Children Playing in the Rain Wet Clothes Taking Longer to Dry Tourists Choosing Hot Beaches Children Using Mosquito Nets More in Rainy Season

						Packing for a Trip Farming Season House Construction Water Availability Flooded Roads

						Umbrella Habit Fishing Restrictions Clothing Choice Travel Decision

4.	Defines a rift valley as a long, wide depression with steep sides. Describes escarpments as the steep sides of a rift valley. Explains that the Great Rift Valley was formed by faulting (breaking of the earth's crust). Identifies the countries crossed by the African Great Rift Valley, from Syria to Mozambique. State that the rift valley is 30–100 km wide.	**iii) DESERT CLIMATE / ARID CLIMATE and (iv) SEMI-DESERT CLIMATE /SEMI-ARID CLIMATE** Desert climate is described as hot and dry throughout the year. Semi-desert climate is described as hot and dry. Desert areas receive very little rainfall because they receive dry winds. Deserts are very hot during day with temperatures ranging between 35º-50ºC. Deserts are very cold at night due to the absence of clouds in the sky. Desert climate is experienced in areas located between the Tropical and Mediterranean climatic region. **Major deserts in Africa.** Sahara desert (the largest and hottest in the world) Namib desert Kalahari desert **African countries that experience desert climate** Libya, Somalia, Namibia, Angola, Western Sahara, Egypt, Morocco **Note:-** <u>Marine deserts</u> are deserts caused by dry winds blowing over large water bodies eg. Namib desert. Continental deserts are deserts caused by dry winds blowing over land for example the Sahara desert, Kalahari desert.	Guided discovery Observation Brainstorming Question and answer Guided group discussion	To help learners understand different environments To help ,earners appreciate diversity of ways of life To help learners understand how people adopt harsh conditions	Identifying Desert climatic region of Africa and their characteristics Drawing the map of Africa showing Desert climatic regions Drawing graphs, charts of tables showing Desert climatic region Explaining the different	Morning Dew Long Lines at Water Wells During the dry season, many people Queue for water at boreholes. Crop Failure Sweating Easily Thick Fog in Early Morning Staying Indoors at Midday Dust Storm Walking Long Distances for Pasture Wearing Sandals

	Mentions economic activities carried out in rift valley areas (crop growing, fishing, tourism, mining, animal rearing). Identifies dangers of rift valleys, such as soil erosion, transport challenges, and high temperatures. Defines a basin as a large low-lying land between highlands and plateaus. Explains that basins are formed by	Harmattan winds are responsible for causing desert conditions in the Sahara. People living in desert areas usually wear light white clothes to reflect sunlight and heat. The desert dwellers build houses with flat roofs in order to prevent desert storms from blowing off their roofs, and also to regulate indoor temperatures. Sand dunes and Oases are the major tourist attractions in desert areas. Sand dunes are large heaps of sand formed by blowing wind in the desert. An oasis is a place where water can be obtained in the desert. The Desert dwellers wear turbans on their heads to keep their bodies cool by preventing the loss of body moisture through the heads **Importance of oases.** They provide water for irrigation farming and for domestic use. They attract tourists who bring in income. **Characteristics of Desert climate.** It is hot and dry throughout the year. It has hot days and cold nights. There is very low humidity. Skies are clear with less cloud cover. It receives very little rainfall.			characteristics of Desert climatic region	Food Shortages Solar Panels Working Better Livestock Drinking More Water Tourists Choosing Hot Beaches Teachers Avoiding Yard Duty in Midday Heat

	down-warping or sinking of land. States that lakes can form in basins when water collects in them. Names examples of major African basins Defines a coastal plain as a narrow strip of land along the coast. Explains that coastal plains support seaport construction and tourism. States that Africa's	**Economic activities commonly done in desert areas.** Irrigation farming, Tourism, Pastoralism **Note:-** Crop farming in desert areas is done around oases through irrigation. Barley, dates and olives are commonly grown in desert areas. Most crops grown in desert areas take a short time to mature (cereal crops) River Nile, lake Chad and oases provide water to dwellers of the Sahara. **Common animals kept in desert areas.** Camels (for use as means of transport), Cattle **Importance of camels to desert dweller** They are used for transport. They are a source of income when sold. They provide meat. They provide milk to them.				Packing for a Trip Farming Season House Construction Seasonal Fruits Water Availability

	coastline is irregular due to wave action. Identify features found on a coastal plain, such as coral reefs, beaches, cliffs, and lagoons. Explains coral reefs as hard sea rocks formed by polyps. Mentions the importance of coral reefs, including tourism, jewellery making, and providing limestone for cement.	**Factors that enable camels to survive in harsh desert conditions.** They have large humps which store fats for so long. They have long eyelashes which protect their eyes from desert sand. They have large hooves which enable them to walk on desert sand. Their nostrils can close which protect which protect their noses from desert sand. Their body temperature changes which prevents water loss from the body through sweating. **A Climatic table showing Desert type of climate.** **A Climatic graph showing Desert type of climate.** **A Climatic table showing Semi-desert type of climate** **A Climatic graph showing Semi-desert type of climate** **Note:-** North Eastern Uganda and the Northern part of Kenya experience Semi-desert climate because they lie on the lee ward side of the Ethiopian highlands.				Desert Travel Drought Umbrella Habit Dusty Roads

						Clothing Choice Travel Decision Water Shortage
5.	Defines a lake as a large depression on the earth's surface where water collects.	**v) MEDITERRANEAN CLIMATE.** It is described as warm dry summers and cool wet winters. It is also called the Warm Temperate Western margin. It is experienced in areas between 30º and 40º North and South of the Equator. It is experienced in areas bordering the Mediterranean sea and the extreme Western corner of South Africa. (Cape region of South Africa) When it is winter in the North, it is summer in the South and vice versa.	Guided discovery Observation Brainstorming	To help learners understand the different climates of Africa To enable learners appreciate diversity of way of life	Identifying **Mediterranean** climatic region of Africa and their characteristics	Morning Dew Long Lines at W During the dry season, many queue for water at boreholes Crop Failure

	Identifies major lakes in Africa, e.g., Lake Victoria, Tanganyika, Malawi, Chad, etc. Explains the different types of lake formations, such as basin lakes, crater lakes, man-made lakes, and ox-bow lakes.	The coldest months in North Africa are December and January while June and July are the hottest. The coldest months in South Africa are June and July while December and January are the hottest. The Westerly winds cause rainfall in the Mediterranean climatic region during winter **African countries that experience Mediterranean climate.** Morocco, Algeria, Tunisia, Libya **Characteristics of Mediterranean climate** It has cool and wet winters It has hot and dry summers. It receives moderate rainfall especially during winter. It experiences high temperatures during summer. Rainfall decreases with the increase in temperature. **A Climatic table showing Mediterranean type of climate** **A climatic graph showing the Mediterranean type of climate** **Note:-**	Question and answer Guided group discussion Picture interpretation	To enable the learners understand how to compare different climatic regions	Drawing the map of Africa showing **Mediterranean** climatic regions Drawing graphs, charts of tables showing **Mediterranean** climatic region Explaining the different characteristics of **Mediterranean** climatic region	Sweating Easily Thick Fog in Early Morning Staying Indoors at Midday Walking Long Distances for Po Wearing Sandals Putting on C Food Shortages Children Playing in the Rain Solar Panels Working Better Wet Clothes Taking Longer to

	States characteristics of basin lakes, including being wide, shallow, fresh-water, and having inlet and outlet rivers. Describes crater lakes as lakes formed on volcanic	According to the above table and the graph, rainfall decreases with the increase in temperature in the Mediterranean climatic region. **Common economic activities carried out in Mediterranean climatic region.** Citrus fruit growing, Lumbering, Animal rearing (pastoralism), Tourism, Wine making **Major crops grown in Mediterranean climatic region.** Citrus fruits like; oranges, lemons, limes, tangerines, grape vines. **Products obtained from citrus fruits** Juice, Wine				Livestock Drinking More Wate Tourists Choosing Hot Beache Children Using Mosquito Nets More in Rainy Season Packing for a Trip Farming Season

	mountains due to volcanicity. Explains man-made lakes as lakes created when dams are built across rivers. Describes ox-bow lakes as lakes formed when meanders are cut off from a river.					House Construction Seasonal Fruits Water Availability Fog in the Highlands Flooded Roads Desert Travel

	Gives examples of each type of lake formation in Africa (e.g., L. Katwe – crater; L. Volta – man-made). Explains why Lake Victoria is an interterritorial lake (shared by three East African countries).					Lake Breeze Drought Snow on Mountain Tops Umbrella Habit Dusty Roads Clothing Choice

	Recognises that some lakes promote trade, fishing, HEP and tourism.					Water Shortage Comparing Two African Regions
6.	Defines a river as a mass of flowing water on the earth's surface. Identifies major rivers in Africa, such as River Nile, Congo, Zambezi, Volta, Senegal,	**vi) TEMPERATE CLIMATE.** Temperate climate is described as warm wet summers and cool dry winters. It is experienced in Eastern parts of South Africa. It extends to the Drakensberg mountains. The areas that experience Temperate climate in South Africa are called the <u>Veld / High Veld</u>. The High Veld is warm with temperatures ranging between 10º-19ºC. **African countries that experience Temperate climate** South Africa, Eswatini, Lesotho **States in South Africa that experience Temperate climate.**	Guided discovery Observation Brainstorming Question and answer	To help learners understand the different climates of Africa To enable learners appreciate diversity of way of life To enable the learners understand how	Identifying temperat e climatic region of Africa and their characteri stics Drawing the map of Africa showing temperat	Morning Dew Long Lines at Water Wells During the dry season, many people queue for water at boreholes. Crop Failure Thick Fog in Early Morning

	Gambia, Limpopo, Rufiji, etc. States that most rivers in Africa originate from highlands and plateaus because they receive a lot of rainfall. Explains that rivers flow from	Orange free state, Cape colony, Natal province **Characteristics of Temperate climate.** It receives moderate rainfall during summer. Rainfall increases with the increase in temperature. It has warm and wet summers. It has cool and dry winters. **A Climatic table showing Temperate type of climate.** **A Climatic graph showing the Temperate type of climate** **Note:-** According to the above table and the graph, rainfall decreases with the decrease in temperature and viceversa. **Economic activities commonly done in temperate climatic region** Sheep rearing, Ranching, Crop cultivation, Dairy farming, Tourism **Note:-** Sheep rearing is the main economic activity carried out in Temperate climatic region (the Veld) Merino sheep is mainly reared in Temperate climatic region. (Mainly for wool production and for mutton)	Guided group discussion Picture interpretation	to compare different climatic regions To help learners to understand how temperate climate influences human activities	e climatic region Drawing graphs, charts of tables showing temperat e climatic region Explaining the different characteri stics of temperat e climatic region	Thunderstorms Children Playing in the Rain Children Using Mosquito

	high altitude to low altitude. Describes sources of river water (rainfall, melting snow, underground water). Explain meanings of key river terms Describes features formed by rivers, such as waterfalls, deltas,	Dairy farming in Temperate region is favoured by the presence of plenty of pasture for animals, and favourable cool climate which favours dairy farming. Crops commonly grown in Temperate climatic region include; maize, sugarcanes, potatoes, wheat, etc.				Nets More in Rainy Season Packing for a Trip Farming Season House Construction Water Availability Fog in the Highlands

	estuaries, gorges, flood plains and ox-bow lakes. Identifies that deltas are formed by river deposition at the river mouth.					Flooded Roads Snow on Mountain Tops Umbrella Habit Clothing Choice

	Describes the three stages of a river States the importance of waterfalls (tourism, HEP) and their dangers (water accidents, block transport).					Travel Decision
7.	States that River Nile is the	**vii) MOUNTAIN CLIMATIC REGION / MONTANE CLIMATIC REGION** Montane climate is experienced in highland areas.	Guided discovery	To help learners understand the	Identifying montane	Morning Dew

	longest river in the world. Identifies the local name of River Nile as "Kiira." Explains why River Nile flows northwards (because the north is at a lower altitude). Identifies Lake Victoria as the source and the Mediterranean Sea as the mouth of River Nile. Names the countries drained by River Nile:	Temperature reduces with the increase in altitude. Temperature reduces by 1ºC for every 100 metres above the sea level. The highest peaks of some mountains in Africa are snowcapped because they rise high above the snow line. The wind ward side receives plenty of rainfall because it receives warm moist air. The lee ward side receives very little rainfall because it receives dry winds. **Some areas in Africa that experience Montane climate.** Areas around mountain Rwenzori Areas around mountain Kilimanjaro Areas around mountain Kenya The Ethiopian highlands Areas around mountain Elgon **African countries that experience Montane climate** Ethiopia, Kenya, South Africa, Uganda **Characteristics of Montane climate.** It has cool temperature. Relief rainfall is mostly received. The wind ward side receives more rainfall than the lee ward side. **Economic activities commonly done in Montane climatic region**	Observation Brainstorming Question and answer Guided group discussion Picture interpretation	different climates of Africa To enable learners appreciate diversity of way of life To enable the learners understand how to compare different climatic regions To enable the learner differentiate between temperature and altitude	climatic region of Africa and their characteristics Drawing the map of Africa showing climatic regions Drawing graphs, charts of tables showing montane climatic region Explaining the different characteristics of montane	Crop Failure Sweating Easily Thick Fog in Early Morning Thunderstorms Flooded Gardens Food Shortages Children Playing in the Rain Solar Panels Working Better

	List the main tributaries of River Nile Identifies major waterfalls on River Nile Names major hydropower dams along the Nile Explains that Aswan High Dam led to the creation of Lake Nasser.	Crop cultivation (Arabica coffee cultivation) Lumbering, Tourism, Dairy farming			climatic region	Wet Clothes Taking Longer to Dry Children Using Mosquito Nets More in Rainy Season Packing for a Trip Farming Season House Construction

	Explains why the Nile is called the "lifeblood of Egypt." Identifies Nile Valley countries and explain that the Nile Valley is the land drained by the Nile and its tributaries. Explains why the Nile Valley is densely populated (fertile soils & reliable water supply).					Water Availability Fog in the Highlands Flooded Roads Snow on Mountain Tops Umbrella Habit

						Fishing Restrictions Clothing Choice Travel Decision
8.	Compares factors that influence the climate of east Africa and that of other regions Explains how climate influences human activities	**FACTORS THAT INFLUENCE THE CLIMATE OF AFRICA** Nearness to water bodies (drainage), Altitude, Prevailing winds, Human activities, Vegetation, Latitude, Ocean currents **The Influence of the above factors on Climate.**	Guided discovery Observation Brainstorming	To help learners understand the different climates of Africa To enable learners appreciate diversity of way of life To enable the learners	Comparing factors that influence the climate of East Africa and that of other regions	Morning Dew Long Lines at Water Wells During the dry season, many people queue for water at boreholes. Crop Failure

	Describes how human activities affect climate		Question and answer Guided group discussion Picture interpretation	understand how to compare different climatic regions To enable the learner differentiate between temperature and altitude	Explaining how climate influences human activities Describing how human activities affect climate	Sweating Easily Thick Fog in Early Morning Staying Indoors at Midday Thunderstorms Dust Storm Flooded Gardens Walking Long Distances for Pasture Wearing Sandals Putting on Closed Shoes Food Shortages Children Playing in the Rain

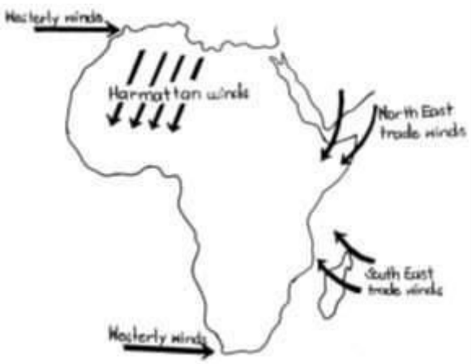
						Solar Panels Working Better Wet Clothes Taking Longer to Dry Livestock Drinking More Water Tourists Choosing Hot Beaches Teachers Avoiding Yard Duty in Midday Heat

						Children Using Mosquito Nets More in Rainy Season Packing for a Trip Farming Season House Construction Seasonal Fruits Water Availability

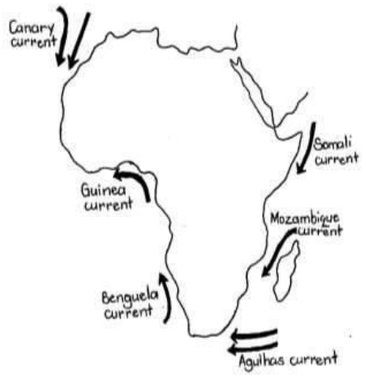
						Fog in the Highlands Flooded Roads Desert Travel Lake Breeze Drought Snow on Mountain Tops Umbrella Habit

						Dusty Roads Fishing Restrictions Clothing Choice Travel Decision Water Shortage Comparing Two African Regions
9.		**PREVAILING WINDS**	Guided discovery	To help the learner	Comparing factors	Clothes drying faster

		Wind is moving air. Wind is caused by the difference in atmospheric pressure. Wind blows from areas of high pressure to areas of low pressure. **Categories of winds.** <u>On shore winds.</u> These blow from the sea to land. They usually bring rain because they pick moisture from seas and lakes over which they blow. <u>Off shore winds</u>. These are winds which blow from land towards the sea. They are usually dry and bring no rain because they pick no moisture from the land over which they blow. **Terms associated with winds**. <u>Trade winds</u> are winds that were used by the early traders to move their boats. ii) <u>Monsoon winds</u> are winds whose direction is reversed from one season to the next. <u>Westerly winds</u> are regular winds that blow from the West towards the East outside the tropics. **Winds that influence the climate of Africa**	Observation Brainstorming Question and answer Guided group discussion Picture interpretation	understand how winds affect weather To help learners understand weather changes To help learners learn about climatic differences	that influence the climate of East Africa and that of other regions Explaining how climate influences human activities Describing how human	Dust blowing in one direction smoke direction sailing boats windmill turning palm trees bending desert sand dunes kite flying smell of food temperature changes swam of insects aeroplane takeoff winnowing seeds

			The North East trade winds	These blow from Arabia towards the Horn of Africa. They bring no rain because they pick no moisture from the desert over which they blow.				activities affect climate	
			The South East trade winds	These blow over the Indian ocean. They bring a lot of rainfall to the Eastern part of Africa because they carry a lot of moisture.					
			The Westerly winds	▪ These bring rain to the Mediterranean areas and Cape province of South Africa.					
			The Harmattan winds.	These are hot and dry winds blowing over the Sahara.					

			They bring no rain to West Africa because they pick no moisture.				
10.	Compares factors that influence the climate of east Africa and that of other regions Explains how climate influences human activities Describes how human activities affect climate	**OCEAN CURRENTS.** **Ocean currents** are masses of water flowing in a particular direction across the ocean. Ocean currents are categorized into warm and cold ocean currents. Warm ocean currents are currents which flow from warm areas to cold areas. For example; Mozambique current, Guinea current, Agulhas current. Cold ocean currents are currents which flow from cold areas to warm areas. For example Somali ocean current, Benguela current, Canary current. **Note:-** The winds that blow over warm ocean current pick a lot moisture and bring rainfall to the main land. The winds that blow over cold ocean currents pick no moisture and bring no rainfall to the main land. **Ocean currents that influence the climate of Africa.**	Guided discovery Observation Brainstorming Question and answer Guided group discussion Picture interpretation	To enable the learner understands climate and weather To help learners learn about ocean life To help learners understand how rainfall distribution is influenced	Comparing factors that influence the climate of East Africa and that of other regions Explaining how climate influences	Cold water at the beach Fishermen catching more fish Fog along the cost Sea shell washed to one side Marine animals migrating Ocean temperature change Sea weed gathering Fishing banned in some months Swimming difficulty



The influence of the above ocean currents on climate of Africa.

Somali current-They cause desert conditions in the Horn of Africa.

Mozambique current-They bring rain to the Eastern side of Madagascar and Southern Africa.

Canary current-They bring no rain to North West Africa.

Benguela current-They cause desert conditions in the Namib and Kalahari desert.

Guinea current-They bring rain to West Africa.

Agulhas current-They cause rain to South Africa.

Note:-

The influence of ocean currents on climate is that areas which receive warm ocean currents have a warmer climate than those that receive cold ocean currents.

Winds which cause different ocean currents.

Winds	Ocean currents caused

human activities

Describing how human activities affect climate

		South East trade winds	• Mozambique currents			
			• Agulhas current			
		South West monsoon winds	• Guinea current			

ALTITUDE

Altitude is the height above the sea level.

Temperature reduces with the increase in altitude.

The lower parts of a mountain are warmer and have higher temperatures than the higher parts of the mountain.

The higher the altitude, the cooler the temperature, and the lower the altitude, the hotter the temperature.

Note:-

Mombasa is hotter than Kasese because Mombasa is at a lower altitude than Kasese.

VEGETATION

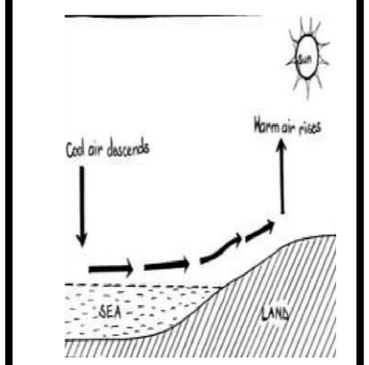
Places with thick vegetation usually receive plenty of rainfall while those that have scanty vegetation usually receive very little rainfall.

Note:-

		Vegetation modifies climate of an area by helping in the formation of convectional rainfall through transpiration. **HUMAN ACTIVITIES** Some human activities affect climate positively while others affect climate of an area negatively. **Human activities that affect climate of an area positively.** Afforestation, Reafforestation, Agro-forestry **Human activities that affect climate of an area negatively.** Deforestation, Swamp drainage, Bush burning, Industrialization				
11.	Compares factors that influence the climate of east Africa and that of other regions Explains how climate influences human activities	**6. LATITUDE** <u>Latitude</u> is the distance in degrees North or South of the equator. Areas which are near the Equator are hotter than those which are far away from the equator. Low latitude areas are areas which are near the equator, and are usually very hot. High latitude areas are areas which are far away from the Equator and are usually cool. Areas which are far away from the Equator are usually very cool because they receive slanting sun rays. **Note;-**	Guided discovery Observation Brainstorming Question and answer	To help learners understand the different climates of Africa To enable learners appreciate diversity of way of life To enable the learners	Comparing factors that influence the climate of East Africa and that of other regions	Morning Dew Hotness at the equator Coldness further from the equator Sunlight strength Crop Failure

	Describes how human activities affect climate	Most parts of Africa are always hot because the sun is overhead Africa throughout the year. The Revolution of the earth (in 365days) causes changes in seasons. ie. Winter, Summer, Spring and Autumn. Tropical region is the area of land lying between the Tropic of cancer and Tropic of Capricorn. Equinox is the time of the year when the sun is overhead the equator. The days and nights are always equal whenever the sun is overhead the equator. Very high temperatures are experienced around equinoxes. Solstice is the time of the year when the sun is overhead the tropics. Areas which are far away from the equator usually experience very low temperatures because they receive slanting sun rays which travel a very long distance to the earth. <u>A diagram showing the time when the sun is overhead major lines of latitude</u> **7) NEARNESS TO WATER BODIES (DRAINAGE SYSTEM)** Areas near large water bodies usually receive more rainfall than those which are far away from large water bodies. Areas near large water bodies usually receive convectional rainfall. Winds that blow over large water bodies pick a lot of moisture which rises and condenses to form convectional rainfall.	Guided group discussion Picture interpretation	understand how to compare different climatic regions To enable the learner differentiate between temperature and altitude	Explaining how climate influences human activities Describing how human activities affect climate	Sweating Easily Thick Fog in Early Morning Staying Indoors at Midday Thunderstorms Dust Storm Flooded Gardens Wearing Sandals Food Shortages Children Playing in the Rain Solar Panels Working Better

		Note:- Water bodies modify climate by helping in the formation of convectional rainfall through evaporation. **Characteristics of convectional rainfall** It is mainly received in the afternoon. It is usually accompanied by lightning and thunder. It is mostly received in areas that experience a lot of evaporation and transpiration. **BREEZES.** A breeze is the movement of air from a cool region to a warm region. **Types of breezes.** Sea breeze, Land breeze **Sea breeze**				Livestock Drinking More Water Teachers Avoiding Yard Duty in Midday Heat Children Using Mosquito Nets More in Rainy Season Packing for a Trip



This is the movement of cool air from the sea towards land.

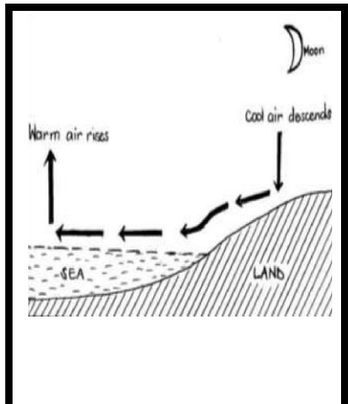
It takes place during the <u>day.</u>

During the day, the land is heated faster than water bodies.

The warm air on land rises and cold air from the sea moves towards land to replace it.

NB:-Sea breezes help in formation of convectional rainfall.

<u>Land breeze</u>



This is the movement of cold air from land towards the sea.

It takes place during the <u>night.</u>

At night, the land cools faster than the sea.

The warm air from the sea where pressure is low rises and the cooler air blows from land towards the sea to replace it.

Farming Season

House Construction

Seasonal Fruits

Water Availability

Fog in the Highlands

Flooded Roads

		PRESSURE BELTS Air pressure / atmospheric pressure is the force exerted by air on the earth's surface. Pressure belts are regions of the earth that experience almost the same atmospheric pressure. Cool areas have high pressure while warm areas have low pressure. Mountain tops and hill tops (highland areas) have high pressure while valleys, plains/ plateaus (low lands) have low pressure. **Causes of difference in pressure.** • Altitude • Latitude • Temperature **Effect of the difference in pressure.** It causes the direction of wind. **Note:-** Wind blows from areas of high pressure to areas of low pressure. Isobars are lines drawn on weather maps connecting places with the same atmospheric pressure. Isotherms are lines drawn on weather maps joining places with equal temperatures.				Lake Breeze Drought Snow on Mountain Tops Umbrella Habit Dusty Roads Fishing Restrictions Clothing Choice

		Isohyets are lines drawn on weather maps connecting places that receive the same amount of rainfall. Isonephs are lines drawn on weather maps connecting places with the same cloud cover. Isohels are lines drawn on weather maps connecting places that receive equal amounts of sunshine. Contours are lines drawn on maps joining places of the same altitude (height above the sea level).				Travel Decision
12.	Compares factors that influence the climate of east Africa and that of other regions Explains how climate influences human activities	**CLIMATE CHANGE** Climate change is the long-term, large scale fluctuation in the patterns of weather experienced in an area. It is the fluctuation of the average weather condition of an area over a long period of time. It is experienced through changes in precipitation, temperature and wind. Global warming is the major cause of climate change. **Global warming** This is the constant rise of world temperatures. Or: This is the continuous heating of the earth due to emission of greenhouse gases to the atmosphere. Examples of greenhouse gases.	Guided discovery Observation Brainstorming Question and answer	To enable learners appreciate lakes as fishing grounds To enable learners understand what is happening in the environment To enable learners recognize trade	Comparing factors that influence the climate of East Africa and that of other regions	

	Describes how human activities affect climate	Carbondioxide, Carbonmonoxide, Nitrous oxide **Causes of global warming** Deforestation Industrialisation, Bush burning, Ozone layer depletion **Note:-** Global warming occurs when the ozone layer is destroyed by excessive sun heat (rays). <u>Ozone layer</u> is a layer of gases that protect the earth from strong sun rays. When trees are cut and bushes burnt, it reduces on the vegetation that would have absorbed carbondioxide which destroys the ozone layer. **Effects / indicators of global warming** Acidic rainfall Melting of snow on high mountains. Excessive heat during day Incidents of severe and prolonged drought. Reduction in the volume of water in water bodies. **Ways of reducing global warming.** By planting more trees. By using renewable sources of energy. eg hydro electricity and solar energy. By treating industrial fumes before emissions.	Guided group discussion	and transport value To enable the learners appreciate lakes and rivers as natural boundaries between countries To help the learner learn about wildlife survival	Explaining how climate influences human activities Describing how human activities affect climate	

		Banning importation of old vehicles. By enforcing laws that protect the environment. **THE INFLUENCE OF CLIMATE ON HUMAN ACTIVITIES** **Desert and Semi-desert climate** **Economic activities commonly carried out in Desert climate.** Pastoralism, Tourism, Crop cultivation. **The influence of desert climate on human activities.** People grow crops by irrigation. People build houses with flat roofs in order to prevent desert storms from blowing off their roofs. Pastoralism is favoured by the presence of large vacant land. Animals with large humps and large hooves are reared eg. camels. Sand dunes and oases encourage tourism. **Problems faced by people living in desert areas.** Sand dust caused by storms. Poor transport High temperatures during day time. Shortage of pasture for animals. Shortage of water. **Possible solutions to the problems faced by people living in desert areas.**				

		By digging valley dams to store water. By practising irrigation farming to increase food supply. By growing drought-resistant crops. By growing fast maturing crops. By rearing camels in order to use them as means of transport. By painting houses with white in order to reflect sunlight. By dressing in light white clothes to reduce heat during day. **b) Equatorial and Tropical climate.** **Economic activities commonly carried out in Equatorial and Tropical climate.** Tourism, Crop growing, lumbering **The influence of Equatorial and Tropical climate on human activities.** Lumbering is done in the tropical rain forests. Tourism is favoured by the plenty of animals. Perennial crops are usually grown due to plenty of rainfall received. People wear thick clothes to provide warmth on rainy and windy weather. People keep a variety of animals due to presence of pasture for animals. (in Tropical climatic region) Houses are built with slanting roofs to allow easy flow of rainwater off the roofs. **Problems faced by people living in Equatorial and Tropical regions.**				

		Floods Attacks from wild animals Poor ground transport Presence of many disease vectors like tsetse flies, mosquitoes **Possible solutions to the problems** **faced by people living equatorial** **and tropical climatic region.** By building houses with slanting roofs. By wearing rain coats and gum boots to safeguard against rainy weather. By creating more wildlife conservation areas. c) <u>Montane climate</u> **Economic activities commonly carried out in Montane climate.** Tourism, Crop growing, Lumbering **The influence of the montane climate on people.** People build houses with slanting roofs. People rear donkeys for use as means of transport. People wear thick clothes to protect their bodies from much coldness. Thick forests promote lumbering in this climatic region. People grow perennial crops due to plenty of rainfall received. **Problems faced by people living in Montane climatic region** Poor transport network.				

		Severe soil erosion Landslides. Much coldness **Possible solutions to the problems faced by people living in montane climatic region** By building winding roads. By keeping donkeys for transport. By wearing thick clothes to overcome much coldness. By planting trees to reduce soil erosion and landslides. **Note:**- Plant roots control landslides in highland areas by holding the soil particles firmly. How does climate affect people's way of dressing? People living in cool areas usually wear thick clothes while those that live in hot areas usually wear light clothes. How does weather affect people's way of dressing? On cold days, people usually wear thick clothes while on hot days, people usually wear light clothes.				